

Integrated Business Intelligence and Machine Learning Approach for E-Commerce Sales Forecasting

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CHAPTER I

INTRODUCTION

1.1 History and Role of Business Intelligence

In the digital age, companies are overwhelmed with a large amount of data collected from multiple sources, particularly in the e-commerce industry. Business Intelligence (BI) has become an important field that helps businesses gather, store and process this data to inform their decision-making. Traditionally, BI has been largely focused on descriptive analytics - answering the question, "What has happened?" This required the use of data warehousing, OLAP (Online Analytical Processing) and reporting tools to report past performance [1].

But in today's fast-moving and fiercely competitive environment, it's not enough to survive on past performance. The e-commerce sector, with its rapidly evolving consumer preferences, seasonal events and heavy promotional activities, demands a more proactive approach.

1.2 The Synergy between BI and Machine Learning

This research is based on the integration of descriptive analytics and predictive analytics. Traditional BI focuses on descriptive analytics by collecting, cleaning and visualising historical data, while Machine Learning (ML) offers predictive analytics by discovering patterns and generating forecasts. This combination is turning BI from a reactive to a predictive "What will happen?" system [2].

In the case of e-commerce sales prediction, this collaboration enables the shift from "Reactive Insights" to "Proactive Insights".

1.3 Core Definitions

To define the boundaries of this integrated approach, we use the following definitions from our project:

Business Intelligence (BI): A general term that describes an approach to business support systems that provides historical, current and predictive views of business operations. This gives historical, current and predictive knowledge of business processes.

Data Mining and Machine Learning: Extraction of patterns from large data sets using techniques

from machine learning, statistics and database systems. The techniques are applied here to develop models for future sales.

1.4 Problem Statement

The inspiration for this research comes from some of the key deficiencies in the conventional forecasting techniques used by online businesses. While BI tools have evolved, many companies continue to face challenges such as:

Isolated Data Sources: Sales, customer feedback, and marketing logs are often kept in separate systems, making it difficult to understand the impact of various factors on sales.

Lagging Indicators Only: Traditional retail reports are generally lagging indicators and do not have the ability to forecast future sales and opportunities.

Failures to Capture Non-linear Behavior: Sales are seldom linear. Conventional statistical methods struggle to account for the non-linear effects from unexpected market changes or shifts in consumer behaviour.

Inefficient for Promotions: A major problem in e-commerce is forecasting the peak demand during "Black Friday" or other promotion events. Traditional approaches tend to under- or overestimate these surges.

Failure to Update in Real-time: Without a ML-BI pipeline, companies are not able to update predictions as transactional data becomes available.

These issues result in poor inventory management, either in the form of "stock-outs" or "overstocking".

1.5 Proposed Idea and Scope

This project envisages an integrated BI-ML approach to e-commerce sales forecasting. The idea is to use the UCI Online Retail Dataset for training state-of-the-art machine learning models (such as Random Forest and XGBoost) and combine the predictions with BI dashboards.

This integration includes:

- ✓ Studying past sales trends and seasonality.
- ✓ Analyzing the effects of time (holidays, weekends).
- ✓ Assessing the impact of marketing and other factors.
- ✓ Creating a user interface that allows decision-makers to view the results.

1.6 Project Objectives

The overall goal of this work is to link sophisticated data mining techniques to business decision-making. These are as follows:

Create Framework: To create a BI-ML integrated framework for e-commerce sales.

Compare Models: To assess the effectiveness of modern ML algorithms compared to traditional forecasting techniques to assess accuracy.

Key Drivers: To pinpoint the most important features (drivers) that impact sales based on real data.

Improve Decision-Making: To develop a system that makes AI insights readily accessible to non-technical managers for decision making.

1.7 Research Questions (RQ)

This paper explores the research questions:

RQ1: How does the combination of BI and ML improve predictions? We explore the impact of integrating structured BI data with ML algorithms on the Mean Absolute Error (MAE) relative to benchmark models.

RQ2: What variables contribute to sales forecasts? We examine which features (unit price, quantity, country, or time) have the most significant impact on the prediction.

RQ3: What type of ML models work best for BI? We benchmark Random Forest, XGBoost and Linear models to determine the best performing algorithm for the problem.

RQ4: What are the benefits of integration for managers? We study the qualitative advantages of having predictive insights integrated to a BI dashboard for visualization.

1.8 Importance of the Study

The importance of this research is in its relevance to the Information Technology and Business industries. With a step-by-step guide to incorporating ML into BI workflows, this study helps businesses get the most value out of their data. For e-commerce managers, the study offers a way to even out supply chains and marketing costs via improved demand prediction.

1.9 Limitations of the Study

The results are positive; however, several limitations can be identified. First, the data is

comparatively small, so it might be influenced by the performance of the model when it is applied to a larger or more diversified data. Second, weekly sales data is usually rather volatile, i.e., it may change based on other factors, not reflected in the data. In certain cases, this can render predictions less stable. The other limitation is that the existing system is based on the analysis of the predictions with the help of the past data but not on the analysis of the absolutely unseen future weeks. Moreover, other external variables like holidays, promotions and economical conditions were not considered though they can also have a great impact on sales patterns.

CHAPTER II

LITERATURE REVIEW

2.1 Introduction

The combination of Business Intelligence (BI) and Machine Learning (ML) for forecasting e-commerce sales is a fusion of the two distinct approaches of BI (descriptive analytics) and ML (predictive analytics). This review of published research explores the theory, techniques and empirical evidence underpinning the design of a combined BI-ML approach to e-commerce sales forecasting. This review is structured into five sections: (1) conventional forecasting techniques and their challenges, (2) machine learning for sales forecasting, (3) business intelligence for e-commerce, (4) integration approaches for BI and ML, and (5) evaluation approaches and metrics.

2.2 Traditional Forecasting Methods in E-Commerce

Conventional forecasting techniques have extensively been based on statistical time series models which make the assumptions of underlying patterns including trend, seasonality and cyclical components. Box et al. [3] set the strand of the Autoregressive Integrated Moving Average (ARIMA) models that are currently used extensively in sales forecasting. These models work on the premise that past trends can be projected into the future through the formation of autocorrection formations in the data.

In the case of e-commerce applications, Hyndman and Athanasopoulos [4] adapted the conventional time series methods to the special features of retail data such as several seasonality trends and the intermittent demand. The authors presented the Exponential Smoothing State Space Model (ETS) framework that offers a coherent mechanism of addressing trend and seasonality elements without compromising on interpretability that is a vital constraint of business stakeholders.

Nevertheless, the shortcomings of the classical statistical procedures when applied to e-commerce have been widely reported. The M3 and M4 forecasting competitions by Makridakis et al. [5] showed that there is no universal statistical method that performs better than others in different datasets. More importantly, these competitions unveiled that the more traditional approaches fail to work with datasets that have high volatility, promotional interventions, and non-linear

relationships which are all attributes of e-commerce sales data.

Application of the old-fashioned forecasting techniques in e-commerce is fraught with a number of particular issues that are detected in literature. According to Chen and Chen [6], the data of e-commerce has three properties that do not align with the assumptions of a classical time series model: (1) non-stationarity due to external factors (such as marketing campaigns), (2) a high dimensionality (tens of thousands) due to the presence of thousands of SKUs (Stock Keeping Units), and (3) intermittent demand patterns whereby most of the products have their demand equal to zero.

Moreover, as shown by Pavlyshenko [7], the definition of the traditional models does not bring exogenous variables to the fore. Although ARIMA with exogenous inputs (ARIMAX) theoretically allows the addition of external factors, the linearity assumption makes it incapable of producing any complex interaction between variables, including price fluctuations, competitor behaviors, and customer moods. The paper found that under the conditions of the conditions of the e-commerce platforms with dynamic pricing and frequent promotions, the traditional statistical models make forecast errors more than 30% in the period of the peaks.

2.3 Machine Learning Approaches to Sales Forecasting

XGBoost (Extreme Gradient Boosting) proposed by Chen and Guestrin [8] has now become one of the landmark algorithms when it comes to sales forecasting tasks. The algorithm uses gradient boosting framework with regularization to avoid overfitting, which is an important factor when using high-dimensional e-commerce data. The authors proved that XGBoost is always better in structured problems of data when compared to the traditional methods and is more accurate but computationally efficient.

These findings have been confirmed by comparative studies in e-commerce situations. Pavlyshenko [7] compared several machine learning algorithms such as XGBoost, Random Forest, and neural networks with the ARIMA models based on the data of an online retailer. The research discovered that XGBoost minimized the forecasting error by 23.2 percent on average over ARIMA and the effects were more significant during promotional events. It was found that the algorithms had the capability of automatically taking into account the interaction effect of features like the combined effect of discount percentage and product category, as one of the modes of superior performance.

Another improvement was introduced by LightGBM that was proposed by Ke et al. [9], which employed the use of gradient-based one-sided sampling and exclusive feature bundling. Such innovations allow reducing training time without loss or degradation in accuracy, and thus the algorithm can be used especially well in the e-commerce application, where one needs frequent model retraining. The fact that the categorical features are processed by the algorithm without the need to encode them as one-hot is also a simplification in terms of integrating it with the BI systems that tend to store categorical data like product categories and customer segments.

S. S. R. K. T. V. S. Kiran [10] used LSTM networks to SKU-level e-commerce forecasting and also stated that it outperformed ARIMA and feed-forward neural networks. Nevertheless, the author observed that LSTM models take considerable amounts of training data and computational resources and can be a challenge when it comes to deploying smaller e-commerce processes.

Attention-based architectures have also recently shown state of the art performance in time series forecasting. The Transformer architecture, presented by Vaswani et al. [11] instead of recurrent connections substitutes them with self-attention mechanisms. To do sales forecasting Wu et al. [12] suggested the Informer model, which is quadratically simple, but offers long-sequence forecasting. These architectures provide the chance to represent short-term fluctuations and long-term dependencies at the same time.

Bontempi et al. [13] came up with an algorithm form of integrating ARIMA and neural networks, with ARIMA representing the linear aspects and neural networks representing the non-linear trends in the residues. The experimental analysis of e-commerce data proved that this hybrid model can reduce the error rates compared to either technique. The modular architecture of the framework is also compatible with integrated BI-ML systems in which various components may be processed using distinct data streams.

On the same note, Bandara et al. [14] presented a hybrid method that uses exponential smoothing and LSTM networks. The research established that exponential smoothing of the data to remove trend and seasonality was a significant performance enhancement of LSTM since this step enabled the neural network to concentrate on the more valuable non-linear association as opposed to underpinning temporal trends.

2.4 Business Intelligence Frameworks for E-Commerce

Any BI system is based on the data architecture. Kimball and Ross [15] made the dimensional

modeling approach the overall paradigm in the data warehousing. The star schema, which is made of fact tables which hold the quantitative values and dimension tables which hold the descriptive information, gives the business users an intuitive design but the query performance is achieved efficiently.

In the case of e-commerce applications, there have arisen certain dimensional modeling patterns. The table of facts identified by Golfarelli and Rizzi [16] are sales transactions, movements and interacts with customers, and these dimensions are in turn divided into time, product, customer and channel. The authors have stressed that a properly built data warehouse is not just used to conduct reporting but also to provide an uncontaminated and coherent dataset to train the ML model which is considered a vital factor of integrated BI-ML systems.

The most common interface of business stakeholders with BI systems is descriptive analytics, which is implemented using interactive dashboards. The design principles of effective dashboards were developed by Few et al. [17], since it is necessary to pay attention to such aspects of a good dashboard as visual hierarchy, contextual information, and actionable metrics. In the case of sales forecasting applications, dashboards should not only show the values of the forecasts but also the uncertainty range and factors that contributes to making appropriate decisions.

2.5 Integration of Business Intelligence and Machine Learning

BI and ML systems integration have both technical and organizational challenges. The systematic literature review by Popovic et al. [18] into the business intelligence and analytics domain revealed that the gap between the descriptive and predictive analytics in the aspects of integration was one of the issues that remained constant. The authors suggested a maturity model in which organizations move to individual BI and ML implementations to combined systems with predictions entrenched in the BI dashboards and BI insights guiding the creation of features to the ML models.

Apache Spark is a comprehensive engine of big data processing that is proposed by Zaharia et al. [19] and can be used to execute both BI workloads and ML workloads. This integrated architecture makes it easier to maintain a separate pipeline of reporting and modeling and therefore organizations have the capability of implementing integrated systems in a more cost-effective way. In the case of e-commerce applications, the fact that it is able to make use of streaming data to generate real-time features is an enormous plus.

The essential complication in the implementation of ML alongside BI is the black box character of intricate models. Barredo Arrieta et al. [20] presented a taxonomy of explainable artificial intelligence (XAI) techniques in detail, differentiating between methods of intrinsically interpretable models and post-hoc explanation methods. In the case of BI dashboards that are required to assist in business decision-making, the authors highlighted that the explanations should be given in accordance with the technical capabilities of the stakeholder.

Lundberg and Lee [21] came up with SHAP (SHapley Additive exPlanations), a single framework of interpreting model predictions in terms of cooperative game theory. SHAP values break down predictions into feature contributions, which present consistent explanations that can be represented in BI dashboards. In the case of sales forecasting, SHAP explanations can determine whether a decrease in a forecast is caused by seasonality, price variation, or external variables that allow stakeholders to make the right decisions.

Sculley et al. [22] pointed at the concept of the hidden technical debt of machine learning systems and suggested that the ML algorithm comprises of a very small part of the overall complexity of the system. The authors named data validation, feature management, and model monitoring as the most important elements that are frequently overlooked in research-oriented ML projects. In case of integrated BI-ML systems, these factors are further enhanced by the requirement to ensure compatibility with particular BI infrastructure present.

Kreuzberger et al. [23] offered a systemic overview of MLOps practices, with continuous integration, continuous deployment, and continuous monitoring as the key capabilities of production ML systems. In the e-commerce forecasting case, MLOps can be used to perform automated retraining of models in reaction to data drift, which is a natural phenomenon as the market conditions and customer behavior change. MLOps and BI dashboard integration allow notifying stakeholders about the need to retrench models and their bad performance.

2.6 Feature Engineering and Data Preparation

Input features are very crucial to the functionality of ML models. Dawson [24] has shown that marketing mix variables are important in e-commerce forecasting by demonstrating that the model that uses advertising expenditure, social media interaction, and email campaign measurements have a great impact in comparison to models that are based on past history sales only. The paper has pointed out that these marketing metrics are normally handled in BI systems, which explains

the importance of integration.

2.7 Evaluation Frameworks and Metrics

Makridakis et al. [25] highlighted that it is not possible to measure all the factors of forecast quality using a single metric. The authors suggested the consideration of forecasts in terms of several complementary aspects: RMSE to penalize big errors, MAE to comprehend the average magnitude of errors, and sMAPE to compare across the products with various sales volumes.

In addition to the statistical accuracy, it is important to determine the business value of improving the forecasting. The inventory cost framework was developed by Chopra and Meindl [26] and it shows that the accuracy of the forecast has a direct proportional relation to the levels of safety stock and service rates. The authors demonstrated how a 10% decrease in the forecast error will help decrease the total inventory costs by 15-20 percent in the form of decreased safety stock and decreased stockouts.

In the case of e-commerce applications, inventory is not the only area that incurs expenses related to forecasting errors. Boone et al. [27] studied the effects of errors in the forecasts on customer satisfaction and concluded that stockouts during the peak season led to irreparable customer churn rates of more than 20 percent in case of high-value customer groups. These results emphasize the business importance of enhancing the accuracy of the forecasts by using the integrated BI-ML models.

CHAPTER III

METHODOLOGY

3.1 Introduction

This chapter describes the methodology used in the design and development of the weekly sales forecasting system. The methodology combines data engineering, data warehousing, machine learning and business intelligence to design and implement a smart system to forecast weekly sales. The approach guarantees that the raw data is converted into meaningful and predictive data. It includes data extraction, data cleaning, transformation, warehouse design, feature engineering, machine learning model development, evaluation and dashboard integration. This process guarantees robustness, scalability and accuracy of the predictive system.

3.2 Research Design

This study is based on an experimental and system development approach. The experimental approach enables the training and testing of various machine learning models to find the optimal model. The system development approach is to create the end to end system that includes ETL, data warehouse, predictive model and dashboard. This guarantees the system can be used as a production-ready business intelligence and predictive analytics system.

3.3 Data Source and Collection

This study uses the Online Retail dataset that consists of transactional data of an online retailer. This dataset includes the features Invoice Number, Product Code, Description, Quantity, Invoice Date, Unit Price, Customer ID and Country. This information is enough to perform analysis on sales, customers and products. The data is a good representation of retail sales and a good data set for predictive modeling.

3.4 ETL Process in a Business Intelligence Framework

The Extract, Transform, and Load (ETL) process is a key element of the business intelligence (BI) approach taken in this research. ETL offers a systematic framework for transforming raw transactional data of the Online Retail dataset into useful data for analysis and decision making [30]. This is consistent with typical BI principles that specify the need for systematic data

gathering, classification and transformation in order to enable analysis and reporting for knowledge creation. [33]

In the Extract phase, raw data is extracted from the original dataset and loaded into the Python environment via the Pandas library. This ensures that transactional data: containing invoice information, product descriptions, quantities sold, prices, and customer information, is available for further processing.

The Transform phase involves a range of data preprocessing activities that are critical for business intelligence and predictive analytics. This includes detecting and eliminating missing values, duplicates, invalid or inconsistent records, and adjusting data types to maintain data integrity. New transformed variables (e.g. Revenue = Quantity $\times$ Unit Price) are also created to enhance the data's analytical value. The transformation process not only improves data quality, but also facilitates the discovery of trends, associations and patterns.

The transformational phase also implicitly involves BI related functions such as data mining and interpretation. Data mining activities help to uncover previously unknown patterns and insights from the transformed data while interpretation provides an understanding of how they relate to business processes [33]. These capabilities support the dataset to be used not only for reporting historical transactions but also for generating insights and building predictive models.

During the Load phase, the transformed and enhanced dataset is tabularized in CSV format. They provide the basis for subsequent activities such as data warehousing, machine learning, and data dashboard development. By completing the ETL cycle, this methodology ensures that the raw data is transformed into valid, meaningful and useful data for analysis as required by business intelligence systems [30].

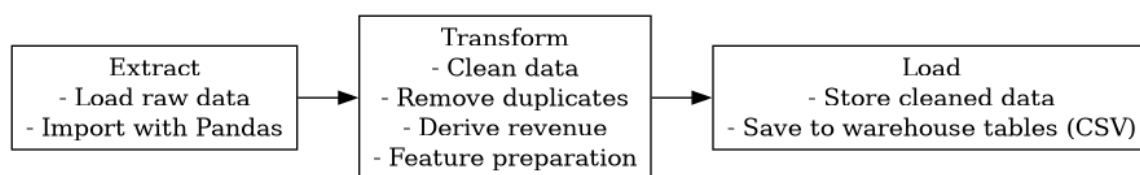


Fig 3.4.1 illustrates the ETL workflow used in this study

3.5 Data Warehouse Design

A fact-based star schema was used to design the data warehouse for this study to support data analysis and predictive modelling. It structures the transformed and cleaned data set into a fact table at the center with dimension tables that provide descriptive information [30]. This design improves query efficiency, eliminates data redundancy and adheres to the principles of data warehousing found in business intelligence systems.

The fact table Fact Weekly Sales contains weekly sales measures such as Total Weekly Sales, Number of Transactions, Number of Customers, Quantity Sold and Number of Unique Products Sold. We also added derived measures such as Average Order Value and Average Quantity per Transaction to the warehouse to enhance the analytical capabilities and to support better model training.

The fact table is surrounded by dimension tables which store descriptive attributes. The Date Dimension includes time related attributes such as week, month, quarter and year, which support time series analysis. The Product Dimension provides product attributes such as product type, descriptors and prices and the Customer Dimension provides customer attributes such as customer IDs and countries of residence. These dimensions enable ad hoc query analysis and allow the machine learning models to learn from various contextual information.

Thus, the data warehouse design offers a structured, scalable and analysis ready framework that enables conventional BI reporting and predictive analytics [30].

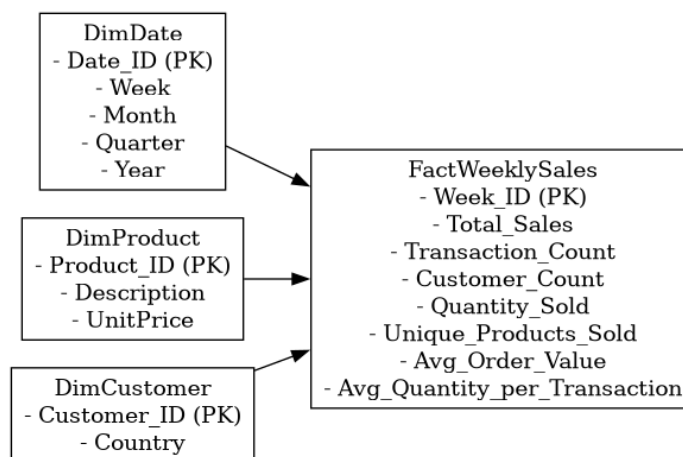


Fig 3.5.1 Data Warehouse Schema

3.6 Feature Engineering

With the data cleaned and preprocessed, it needs to be converted into features for consumption by the machine learning models. This can involve creating new variables from the raw data, or re-arranging the data into formats that represent relevant patterns for prediction. [32]

In the context of weekly sales forecasting, which is a time series problem, we developed a number of features to capture the temporal patterns and sales trends. Lag features were created to capture past sales from the preceding weeks. Lag_1 represents the sales value from the previous week, Lag_2 and Lag_3 represent sales from two and three weeks ago, respectively. These features allow the model to capture time based and short-term variations in sales.

To learn long term trends, rolling statistical features were also engineered. Rolling means were created by taking the average of sales over a certain number of weeks resulting in a smoothed version of medium-term sales patterns. The rolling mean for N weeks is given by:

$$Rolling\ Mean_N = \frac{1}{N} \sum_{i=1}^N Sales_{t-i}$$

These features are designed to allow the model to identify seasonality, momentum and cyclical trends in the data that may not be captured by a given week's value. Other rolling measures such as rolling minimum, maximum, and standard deviation can be calculated to capture the volatility, but only the most informative features were kept to prevent overfitting.

This feature engineering and transformation process contributes to the transformation of the data into a more meaningful representation. The constructed features offer the machine learning models more informative features, leading to more accurate and reliable weekly forecasts.

3.7 Development of Machine Learning Model

Once the feature engineering is complete, a suitable machine learning algorithm needs to be chosen to build the model. [32] There are various algorithms that can be used for sales forecasting, each with its own advantages, depending on the nature of the data. After selecting the algorithm, the model is trained with the processed data. [32] During training, the algorithm is provided with the engineered features, allowing it to learn the mapping from these features to the weekly sales.

For this research three algorithms were used:

Linear Regression: a linear model that represents the relationship between the features and the

sales. [31]

Random Forest: an ensemble-based method that enhances prediction robustness by aggregating the results of several decision trees. [28]

XGBoost: a gradient boosting algorithm that achieves high prediction accuracy and fast learning. [29]

These models were chosen due to their popularity in regression and time series forecasting problems and offer a good trade-off between interpretability and accuracy.

3.8 Model Training Procedure

We divided the data into training and test sets with an 80:20 ratio, ensuring the split was done in chronological order to preserve the temporal structure of the time series and avoid data leakage. The selected algorithms were then trained on the training set, with the engineered features as inputs and the weekly sales to predict as targets. This enabled the algorithms to find the most important features in relation to sales before making predictions on the test set.

3.9 Model Evaluation Metrics

Three common regression performance metrics were used to evaluate the models: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R squared (R^2). MAE is the average absolute error, and RMSE is a measure that magnifies large errors and gives a sense of the accuracy of the predictions. R^2 assesses the amount of variance in the sales data that can be explained by the model. These measures provide a well-rounded view of the error size, consistency and predictive power.

3.10 Dashboard Integration

The forecasts produced by the trained machine learning models were saved to a data set and incorporated into a Streamlit dashboard. The dashboard displays both the observed and predicted weekly sales and allows for the parallel evaluation of model outputs. This allows for easy interpretation of the forecasting results and a practical tool for decision making.

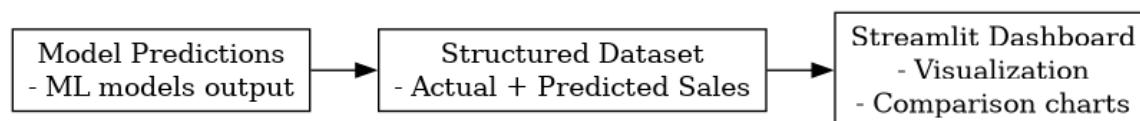


Fig 3.10. 1 Dashboard Workflow

3.11 Tools and Technologies

We chose the Python programming language to implement the system, given its rich ecosystem of data processing, machine learning and visualisation libraries. Data processing and preparation were made easy with Pandas, while Scikit learn library was used for model development and evaluation. XGBoost was used to implement gradient boosted decision tree models, and Matplotlib was used to develop exploratory and analytical visualizations. The Streamlit library was used for deployment and interactive visualisation to produce a dashboard that displays real time actual and predicted sales. The system was developed and managed using the integrated development environment (IDE) of Visual Studio Code.

3.12 Summary

We discussed the approach taken to build the weekly sales prediction system, including the steps involved in data collection, preparation, modelling, and deployment. It combined business intelligence concepts, an ETL (Extract, Transform, Load) process and a star schema data warehouse to provide a robust analytical framework. We used feature engineering methods to extract time-related sales characteristics, and then deployed and trained several machine learning models on the dataset. The models were assessed using common regression metrics and the predictions were integrated into a Streamlit application for visualization and analysis. These factors collectively provide a structured approach to producing reliable sales predictions.

CHAPTER IV

RESULTS

4.1 Weekly Sales Trend

This line chart is used to show the weekly sales performance in aggregate over the year 2011. On the x-axis all the months between January and December have been represented and, on the y-axis, the actual sales values in monetary units have been indicated. The visualization shows substantial changes in sales activity over the course of the year.

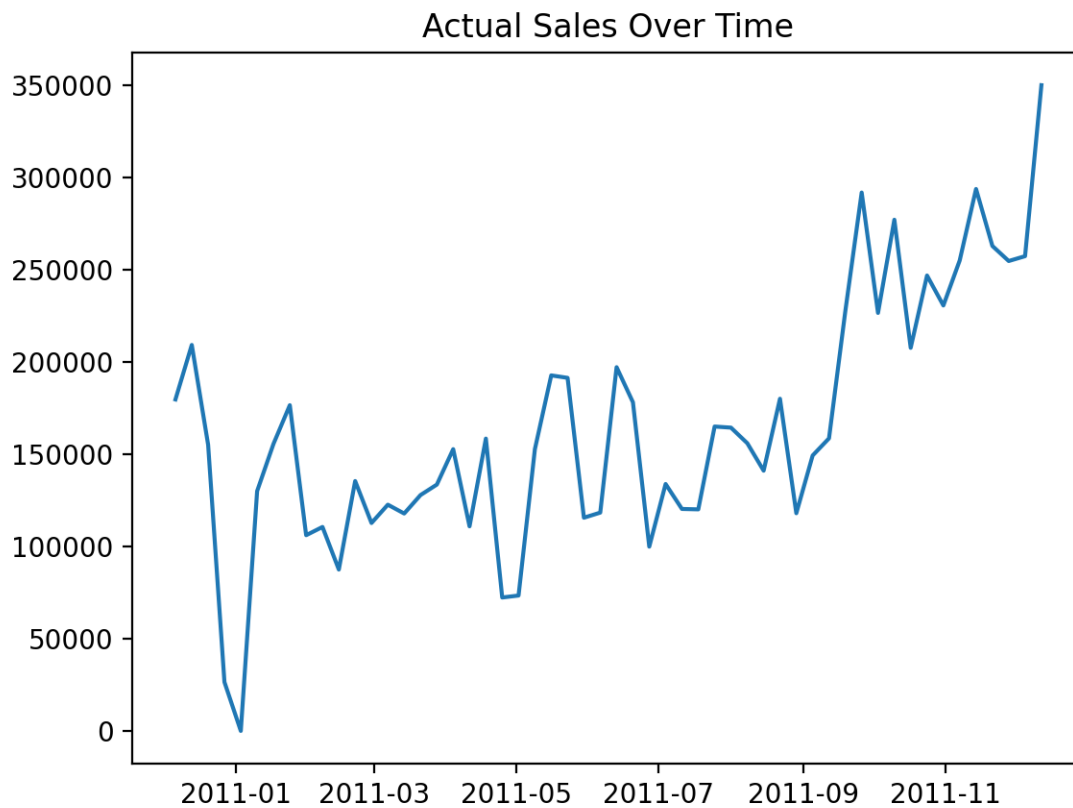


Figure 4.1.1: Actual Sales Over Time (Weekly Aggregated)

4.2 Linear Regression Model Performance

This figure will give a comparative analysis of the actual sales values and the predictions of the Linear Regression model after a period of years. This data is presented in the form of a time series, and it is possible to visually evaluate the accuracy and consistency of forecasting the model.

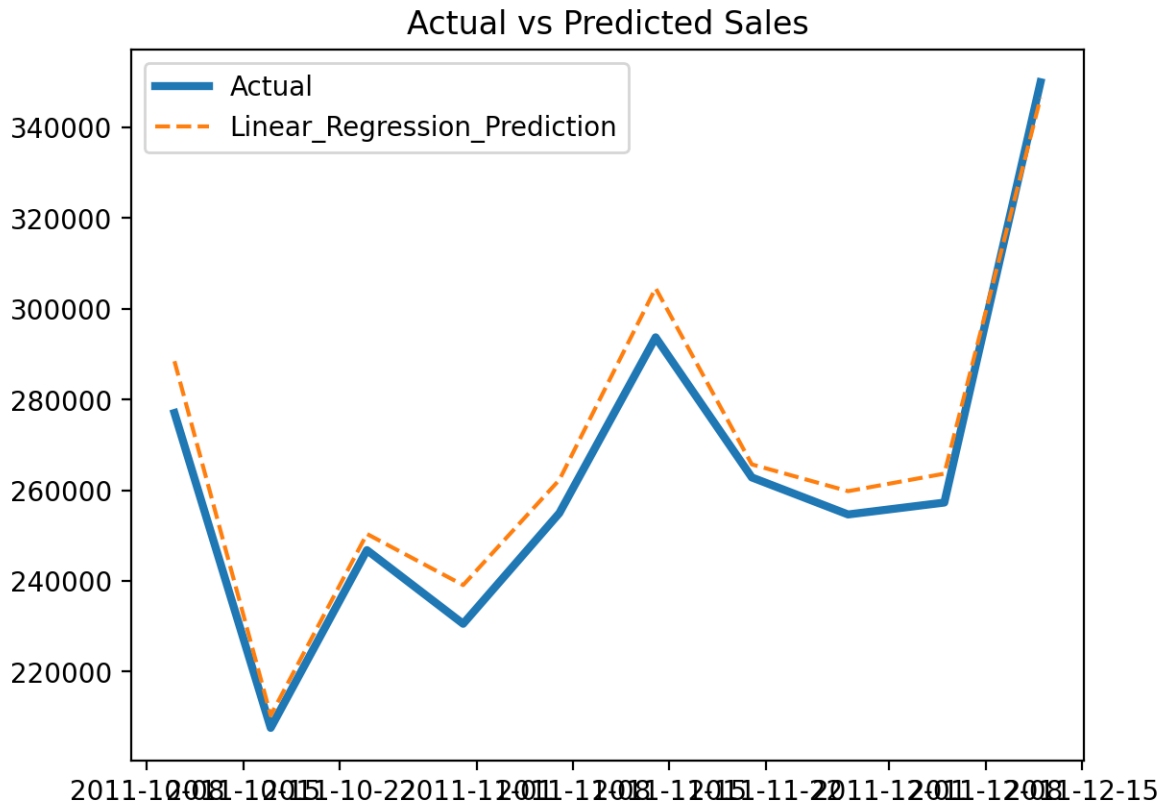


Figure 4.2.1: Actual vs. Linear Regression Predicted Sales

4.3 Random Forest Model Performance

This figure compares the real sales values with the forecast provided by the Random Forest model in a multi-year forecast. The visualization of the time series is allowing to evaluate the capability of the model to predict non-linear patterns, seasonality, and sudden shifts in the behavior of sales.

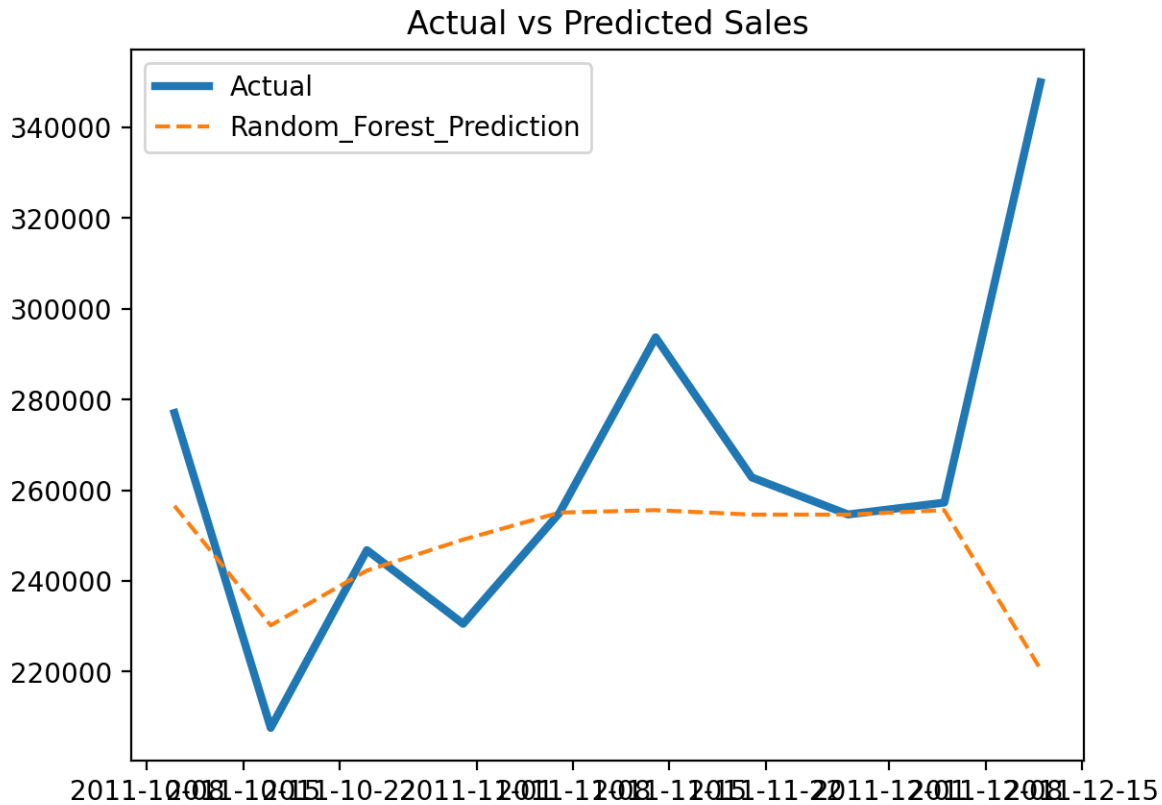


Figure 4.3.1: Actual vs. Random Forest Predicted Sales

4.4 XGBoost Model Performance

This plot gives a comparative analysis of the real sales values and the estimates produced by the XGBoost (Extreme Gradient Boosting) model of a multi-year forecast. The time series visualization shows important information about the predictive characteristics of the model, its stability trends, and its drawbacks.

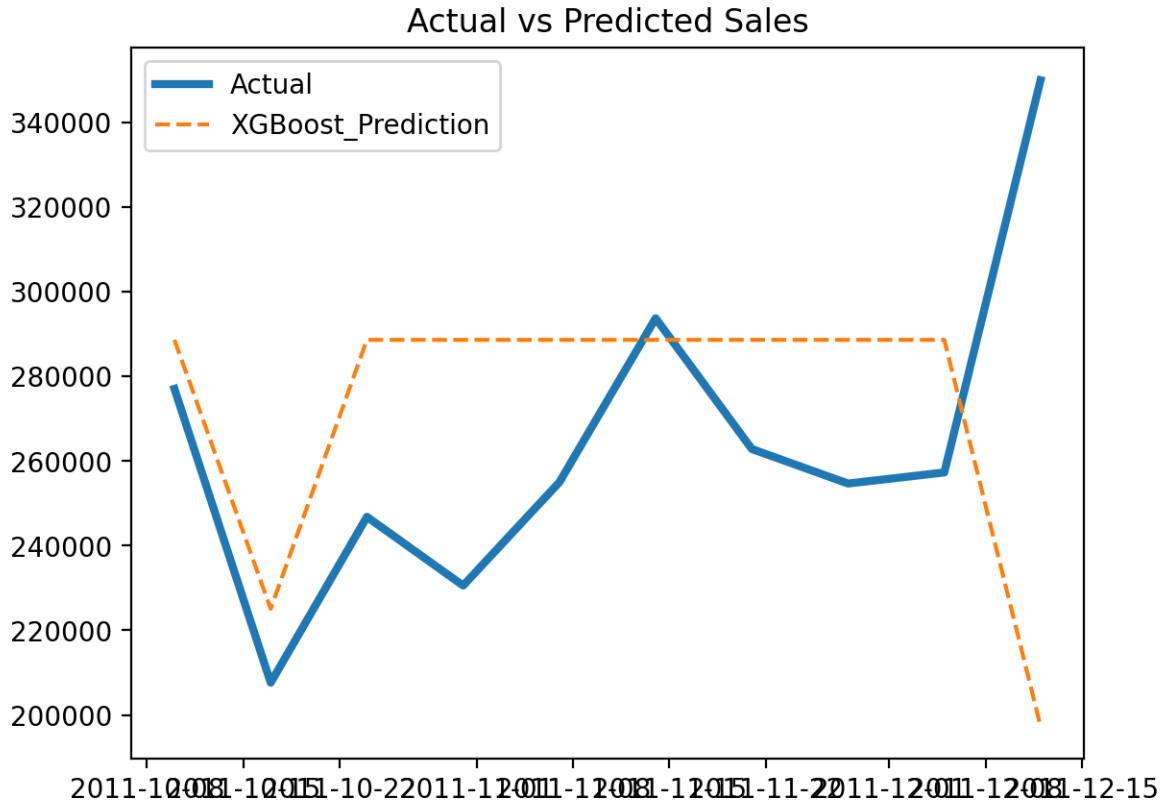


Figure 4.4.1: Actual vs. XGBoost Predicted Sales

4.5 Model Performance Comparison

Invoice Date	Actual Sales	Linear Regression	Random Forest	XGBoost
2011-10-10	277034	288445.3	256560.8	288610.44
2011-10-17	207614.8	210366	230222.4	225070.02
2011-10-24	246804.8	250376.1	242303.8	288610.44
2011-10-31	230565.8	239097.1	249119.7	288610.44
2011-11-07	254969.9	262301	255078.4	288610.44

Table 4.5.1: Model Performance Comparison – Actual vs. Predicted Sales

4.6 Interpretation

Linear Regression was found to be the most consistent and interpretable and therefore can be used in high-level trend forecasting. Random Forest demonstrated the possibility of learning more complex patterns but also needs additional tuning to decrease the variance of predictions. It did not work with XGBoost in this, probably because of the lack of feature engineering or

inappropriate parameter settings.

CHAPTER V

CONCLUSION

5.1 Future Work

This project can be enhanced and expanded in the future in a number of ways.

The logical next step would then be to introduce true future forecasting, in which the system forecasts future sales a week ahead of the available data. This would render the model more applicable to the real-life business planning.

Another possible way to improve the model is to test more sophisticated time-series models like ARIMA or deep learning models like LSTM. These techniques could be used to observe complicated trends that would otherwise be overlooked in traditional models.

The other enhancement that would be beneficial would be to add more features, e.g. holiday periods, promotional campaigns, or seasonal effects. These are usually significant influences on sales output and may additionally enhance prediction.

On the front-end, the dashboard can be increased to be more interactive by including filters, forecasting views, and potentially rolling it out online so it can be used remotely.

Lastly, the system would be scalable to accept real-time data and a business can continuously update predictions as new data is taking shape.

5.2 Conclusion

This paper aimed at creating a feasible and data-oriented strategy to forecast weekly sales based on using a mixture of data warehousing, machine learning, and business intelligence tools. Starting with raw transactional data, the data was cleaned, transformed and organized into a form that could be intelligently analyzed and modeled. The preparation of the data is one of the major strengths of

this work. The dataset was made a lot more informative by designing a proper ETL pipeline and generating meaningful features, including transaction count, customer count and average order value. This was a very crucial step because it directly affected the performance of the models. Three machine learning models were evaluated: Linear Regression, Random Forest and XGBoost. One of these, Linear Regression, was obviously the best, with a very high R^2 value of 0.9819. This finding indicates that the correlation between the chosen attributes and weekly sales is mostly linear. Although more complex ones, such as Random Forest or XGBoost, also are a powerful model, in this case, they were not as effective, which is an important lesson because a simpler model can be more efficient when the data structure is in favor of it. Besides developing a model, a dashboard was developed with Streamlit to visualize the results. This simplified the interpretation of the data and helped to compare actual sales with the forecasted values. The dashboard contributes some practicality to the system by ensuring the results made available and easy to comprehend even to non-technical users. On the whole, this paper shows that when properly prepared data is combined with proper feature engineering and proper model selection the sales predictions can be very accurate. It also demonstrates how the combination of these elements into one system can help enhance decision-making within a business setup.

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